

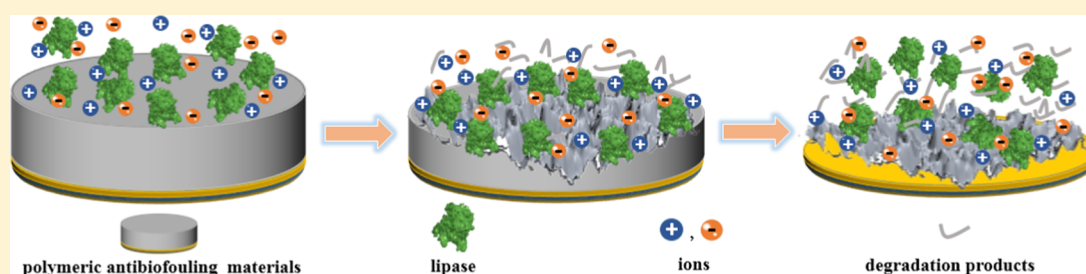
Specific Ion Effects on the Enzymatic Degradation of Polymeric Marine Antibiofouling Materials

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Supporting Information



ABSTRACT: It is expected that the widely dispersed ions in seawater would have strong influence on the performance of polymeric marine antibiofouling materials through the modulation of enzymatic degradation of the materials. In this work, poly(ϵ -caprolactone)-based polyurethane and poly(triisopropylsilyl methacrylate-*co*-2-methylene-1,3-dioxepane) have been employed as model systems to explore the specific ion effects on the enzymatic degradation of polymeric marine antibiofouling materials. Our study demonstrates that the specific ion effects on the enzymatic degradation of the polymer films are closely correlated with the ion-specific enzymatic hydrolysis of the ester. In the presence of different cations, the effectiveness of the enzyme to degrade the polymer films is dominated by the direct specific interactions between the cations and the negatively charged enzyme molecules. In the presence of different anions, the kosmotropic anions give rise to a high enzyme activity in the degradation of polymer films induced by the salting-out effect, whereas the chaotropic anions lead to a low enzyme activity in the degradation of the polymer films owing to the salting-in effect. This work highlights the opportunities available for the use of specific ion effects to modulate the enzymatic degradation of polymeric antibiofouling materials in the marine environment.

■ INTRODUCTION

Marine antibiofouling materials have attracted increasing interest because marine biofouling causes many problems in maritime industries.^{1,2} The antibiofouling materials which are capable of releasing toxic molecules have been widely used to effectively inhibit the growth of organisms on various surfaces in the marine environment.^{3,4} However, the release of toxic molecules (e.g., tributyltin) has a strongly negative impact on the marine environment.⁴ Consequently, these kinds of antibiofouling materials have been gradually banned in maritime industries to protect the marine environment.^{4–6} To realize the goal of excellent performance while maintaining the eco-friendly properties of the marine antibiofouling materials, many types of biodegradable polymeric antibiofouling materials have been developed in the past 10 years.^{7–9} The key strategy to achieve the superior antibiofouling performance of the biodegradable polymeric materials is to generate a self-renewing surface after removing the layer of deposited organisms through the degradation of the materials by the attacking of either seawater or enzymes secreted from microorganisms in the marine environment.^{9–12}

It is well known that the marine environment contains a large amount of ions with a total ionic strength of ~ 0.72 M.¹³ The cations are mainly composed of Na^+ , Mg^{2+} , Ca^{2+} , and K^+ , and the anions are mainly composed of Cl^- , SO_4^{2-} , HCO_3^- , and Br^- .¹⁴ On the other hand, it has been reported that the activities of enzymes are strongly dependent on the nature of the ions presented in the systems.^{15–19} Generally, an increase in enzyme activity can be observed in the presence of kosmotropic ions due to an increased affinity between enzyme molecules and substrates through the salting-out effect.¹⁵ In contrast, the chaotropic ions can denature the enzyme conformation via the salting-in effect, leading to a decrease in enzyme activity.¹⁵ Therefore, it is expected that the enzymatic degradation of polymeric antibiofouling materials should be influenced by the nature of the ions presented in the marine environment. No doubt, understanding of the specific ion effects on the enzymatic degradation of polymeric

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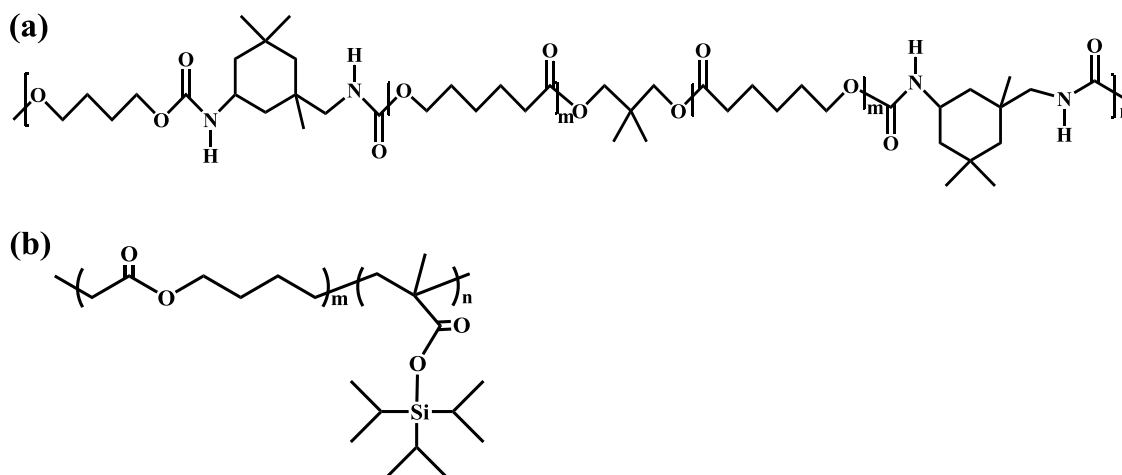


Figure 1. Chemical structures of PCLPU (a) and PTIO (b).

antibiofouling materials is important for improving the performance of the materials in the marine environment.

In this work, two types of polymers, i.e., poly(ϵ -caprolactone)-based polyurethane (PCLPU) and poly-(triisopropylsilyl methacrylate-*co*-2-methylene-1,3-dioxepane) (PTIO), have been employed as model systems to explore the specific ion effects on the enzymatic degradation of biodegradable polymeric marine antibiofouling materials.^{20,21} The chemical structures of PCLPU and PTIO are shown in Figure 1. As can be seen from Figure 1, PCLPU has a degradable main chain, whereas PTIO has both the degradable ester part and the hydrolyzable silyl ester part. That is, in comparison to PCLPU, PTIO provides dual sources for self-polishing properties in the marine environment.²¹ These two different polymers are employed here to firmly validate the specific ion effects on the enzymatic degradation of polymeric marine antibiofouling materials. Lipase has been chosen as a model enzyme to degrade the polymeric antibiofouling materials, as lipase is a microbial extracellular enzyme and is widely scattered in nature including the marine environment.²² We have found that the enzymatic degradation of the polymeric antibiofouling materials is strongly affected by the nature of the presented ions, which opens up opportunities for the application of specific ion effects to modulate the biodegradation of polymeric antibiofouling materials in the marine environment.

EXPERIMENTAL SECTION

Materials. Amano lipase PS, from *Burkholderia cepacia*, was purchased from Sigma-Aldrich. *Para*-nitrophenyl butyrate (pNPB, 98%) was purchased from Aladdin. Acetonitrile was purchased from Sinopharm. Sodium chloride (NaCl, 99.99%), potassium chloride (KCl, 99.99%), calcium chloride (CaCl₂, 99.99%), magnesium chloride (MgCl₂·6H₂O, 99.99%), sodium bromide (NaBr, 99.99%), sodium bicarbonate (NaHCO₃, 99.99%), and tris(hydroxymethyl)-aminomethane (Tris, 99.9%) were purchased from Aladdin and used as received. Sodium sulfate (Na₂SO₄, 99%) was purchased from Alfa Aesar and used as received. PCLPU and PTIO were prepared according to the procedures reported previously.^{20,21} The number-average molecular weight (M_n) and polydispersity index (PDI) of PCLPU were $\sim 2.7 \times 10^4$ g/mol and ~ 1.9 , respectively. For the PTIO, the M_n and PDI were $\sim 3.5 \times 10^4$ g/mol and ~ 2.0 , respectively. The solution pH in this work was kept the same as the natural seawater (~ 8.2) using a 5 mM Tris-HCl buffer as the background solution (Figure S1, Supporting Information). The water used was purified by filtration through a Millipore Gradient system after predistillation,

giving a resistivity of 18.2 M Ω cm. Artificial seawater was prepared following ASTM D1141-98 (2003).

Preparation of PCLPU and PTIO Films. The gold-coated substrates were cleaned by piranha solution for ~ 10 min at ~ 60 °C and then successively rinsed with water, dried with nitrogen, and cleaned by plasma treatment at a power of ~ 18 W for ~ 15 min prior to the preparation of PCLPU and PTIO films. The polymer films were prepared by spin casting of polymer/THF solution (6.0 mg/mL) on a spin coater (CHEMAT, KW-4A) at 4000 rpm for ~ 60 s in air. Afterward, the polymer films were heated to ~ 60 °C in an oven for ~ 12 h. The prepared polymer films had a thickness of 43 ± 2 nm, determined using spectroscopic ellipsometry (M2000 V, J. A. Woollam).

Enzymatic Degradation of PCLPU and PTIO Films. Enzymatic degradation of the polymer films was investigated by using a quartz crystal microbalance with dissipation (QCM-D) from Q-sense AB.²³ The quartz crystal resonator with a fundamental resonance frequency of ~ 5 MHz was mounted in a fluid cell with one side exposed to the solution. The resonator had a mass sensitivity constant (C) of 17.7 ng cm⁻² Hz⁻¹.²⁴ When a quartz crystal is excited to oscillate in the thickness shear mode at its fundamental resonance frequency (f_0) by applying an RF voltage across the electrodes near the resonance frequency, a small layer added to the electrodes induces a decrease in resonance frequency (Δf), which is proportional to the mass change (Δm) of the layer. In vacuum or air, if the added layer is rigid, evenly distributed, and much thinner than the crystal, then Δf is related to Δm and the overtone number ($n = 1, 3, 5, \dots$) by the Sauerbrey equation²⁵

$$\Delta m = -\frac{\rho_q l_q \Delta f}{f_0 n} = -C \frac{\Delta f}{n} \quad (1)$$

where f_0 is the fundamental frequency and ρ_q and l_q are the specific density and thickness of the quartz crystal, respectively. The dissipation factor (D) is defined by²³

$$D = \frac{E_d}{2\pi E_s} \quad (2)$$

where E_d is the energy dissipated during one oscillation and E_s is the energy stored in the oscillating system. The measurement of dissipation factor is based on the fact that the voltage over the crystal decays exponentially as a damped sinusoidal when the driving power of a piezoelectric oscillator is switched off.²³ By switching the driving voltage on and off periodically, we can simultaneously obtain a series of changes in resonance frequency and dissipation factor. The enzyme concentration was fixed at 0.5 mg/mL during the degradation of the polymer films. All of the results obtained in this work were from the measurements of frequency and dissipation factor at the third overtone ($n = 3$) and all of the experiments were conducted at ~ 25

°C. The atomic force microscopy (AFM) (Veeco diInnova) measurements of the polymer films were performed using tapping mode in air. The cantilever spring constant is 40 N/m, and the resonance frequency is 300 kHz. The surface morphologies and the root-mean-square (RMS) roughness were obtained from the AFM measurements by using the NanoScope Analysis 1.8 software.

Enzymatic Hydrolysis of pNPB. In this work, pNPB was employed as a model small-molecule substrate to mimic the ion-specific enzymatic hydrolysis of ester. The lipase activity during the enzymatic hydrolysis of pNPB in the presence of different types of ions was determined by monitoring the time-dependent absorbance of the released *para*-nitrophenyl at 405 nm using a microplate reader (SpectraMax M2e). The enzymatic hydrolysis was initiated by adding lipase to the Tris-HCl buffer containing the substrate and salt at ~25 °C. The enzyme concentration was fixed at 0.02 mg/mL, and the substrate concentration was varied from 25 to 1250 μ M for the enzymatic hydrolysis.

RESULTS AND DISCUSSION

In the marine environment, the concentrations of cations are ~468.0, ~54.6, ~10.5, and ~10.2 mM for Na^+ , Mg^{2+} , Ca^{2+} , and K^+ , respectively.²⁶ For the anions in the marine environment, the concentrations of Cl^- , SO_4^{2-} , HCO_3^- , and Br^- are ~545.9, ~28.8, ~2.4, and ~0.9 mM, respectively.²⁶ To clarify the specific ion effects on the enzymatic degradation of the biodegradable polymeric antibiofouling materials in the marine environment, we have investigated the degradation of the polymer films by lipase in the presence of different types of cations or anions with the concentrations the same as those in the natural seawater. Figure 2a shows the time dependence of Δf for the enzymatic degradation of the PCLPU film in the presence of different cations. For all of the cations, the increase in Δf with increasing time is an indication that the PCLPU film can be degraded by the lipase regardless of the nature of the cations. Nevertheless, the cation identity has a significant influence on the rate of enzymatic degradation of the PCLPU film, as reflected by the distinct changes in Δf with time in the presence of different cations. Note that no obvious degradation of the PCLPU film can be observed in both the Tris-HCl buffer and the artificial seawater in the absence of lipase (Figure S2, Supporting Information). According to the change in Δf shown in Figure 2a, the rate of enzymatic degradation of the PCLPU film in the presence of different cations increases following the trend $\text{Na}^+ < \text{Mg}^{2+} < \text{Ca}^{2+} < \text{K}^+$. Figure 2b demonstrates that the change in ΔD for the enzymatic degradation of the PCLPU film is complex in the presence of different cations. ΔD increases first and then gradually decreases with increasing time in the presence of Na^+ . For Mg^{2+} , ΔD has an initially small increase, followed by a decrease in ΔD with increasing time. In the presence of K^+ , ΔD rapidly decreases with increasing time in the initial stage, and then slowly decreases with a further increase in time. For the case of Ca^{2+} , ΔD remains almost constant with increasing time in the whole process of enzymatic degradation.

Generally, Δf is related to the mass change of polymer film on the surface of resonator, whereas ΔD is correlated with the energy dissipated during the oscillation of the resonator.^{27–30} As a consequence, Δf should increase accompanying the degradation of PCLPU film. This is what is exactly observed for the change in Δf during the enzymatic degradation in the presence of different cations, as shown in Figure 2a. For the change in ΔD , the degradation of polymer film would lead to a decrease in ΔD , whereas an increase in surface roughness of the polymer film would result in an increase in ΔD due to an

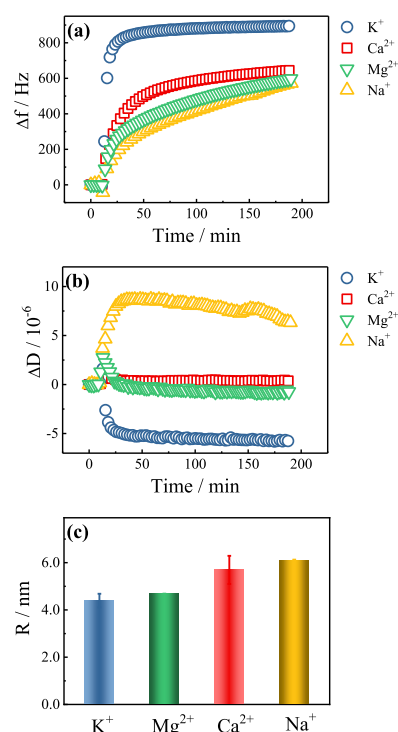


Figure 2. (a) Shift in frequency (Δf) for the enzymatic degradation of the PCLPU film as a function of time in the presence of different cations with Cl^- as the common anion. (b) Shift in dissipation (ΔD) for the enzymatic degradation of the PCLPU film as a function of time in the presence of different cations with Cl^- as the common anion. (c) RMS surface roughness (R) of the PCLPU film after a certain time of enzymatic degradation in the presence of different cations with Cl^- as the common anion. For K^+ , Ca^{2+} , and Mg^{2+} , the surface roughness is determined at the time of enzymatic degradation of ~5 min. For Na^+ , the surface roughness is determined at the time of enzymatic degradation of ~25 min, where ΔD has the maximum value. Here, all of the cations in the salt solutions have the same concentrations as those in the natural seawater.

increase in friction between the polymer film and the surrounding water molecules.^{31–36} The AFM results show that the relatively flat PCLPU film becomes rough upon the enzymatic degradation in the Tris-HCl buffer (Figure S3, Supporting Information), leading to an appearance of the maximum ΔD with time during the enzymatic degradation of the PCLPU film (Figure S4, Supporting Information).

In the presence of K^+ , the sharp increase in Δf with increasing time indicates the fast degradation of the PCLPU film. Meanwhile, the rapid decrease in ΔD with increasing time in the presence of K^+ implies that the degradation of the PCLPU film dominates over the increase in surface roughness of the film in terms of their influences on ΔD . This is understandable because the K^+ leads to both the highest rate of enzymatic degradation (Figure 2a) and the lowest surface roughness of the PCLPU film (Figure 2c). For the case of Ca^{2+} , the rate of degradation of the PCLPU film is lower than that for K^+ , as reflected by a relatively slow increase in Δf with increasing time. On the other hand, the surface roughness of the PCLPU film in the presence of Ca^{2+} is larger than that for K^+ . The fact that the ΔD remains almost constant with increasing time in the presence of Ca^{2+} suggests that the influence of the degradation of PCLPU film on ΔD may be offset by the effect generated by the increase in surface

roughness of the PCLPU film during the degradation. In comparison to Ca^{2+} , a slower increase in Δf with increasing time in the presence of either Na^+ or Mg^{2+} is indicative of a slower degradation of the PCLPU film. The initial increase in ΔD with increasing time for either Na^+ or Mg^{2+} indicates that the change in ΔD in this initial stage is dominated by an increase in surface roughness of the PCLPU film owing to the slow degradation. The larger value of the maximum ΔD for Na^+ compared to that for Mg^{2+} is attributed to the larger surface roughness of the PCLPU film for the former than that for the latter (Figure 2c).

In Figure 3a, the time dependence of Δf in the presence of different anions demonstrates that the effectiveness of enzyme

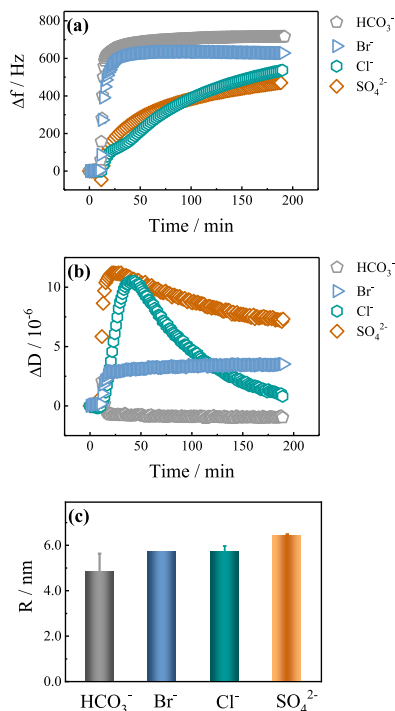


Figure 3. (a) Shift in frequency (Δf) for the enzymatic degradation of the PCLPU film as a function of time in the presence of different anions with Na^+ as the common cation. (b) Shift in dissipation (ΔD) for the enzymatic degradation of the PCLPU film as a function of time in the presence of different anions with Na^+ as the common cation. (c) RMS surface roughness (R) of the PCLPU film after a certain time of enzymatic degradation in the presence of different anions with Na^+ as the common cation. For HCO_3^- and Br^- , the surface roughness is determined at the time of enzymatic degradation of ~ 6 min. For Cl^- and SO_4^{2-} , the surface roughness is determined at the times of enzymatic degradation of ~ 33 and ~ 12 min, respectively, where ΔD has the maximum values. Here, all of the anions in the salt solutions have the same concentrations as those in the natural seawater.

to degrade the PCLPU film increases following the trend $\text{Cl}^- \approx \text{SO}_4^{2-} < \text{Br}^- < \text{HCO}_3^-$. In Figure 3b, ΔD exhibits distinct changes with time in the presence of different anions for the degradation of PCLPU film. For SO_4^{2-} and Cl^- , ΔD increases first and then decreases with increasing time. For the case of Br^- , ΔD increases with increasing time in the initial stage and then remains almost constant with a further increase in time. By contrast, as time increases, an initially rapid decrease followed by a slow decrease in ΔD can be observed for HCO_3^- . The distinct changes in ΔD with time in the presence

of different anions can be understood based on the similar mechanism to that for the cations. For instance, the combination of the high rate of degradation and the low surface roughness in the presence of HCO_3^- gives rise to a rapid decrease in ΔD with increasing time, whereas the combination of the low rate of degradation and the high surface roughness in the presence of either Cl^- or SO_4^{2-} leads to the appearance of the maximum ΔD with time (Figure 3c).

During the degradation of PCLPU film, the enzyme concentration is fixed at 0.5 mg/mL. According to the previous studies, the degradation of PCLPU film could be described using a pseudo-first-order reaction equation.^{37–39} As a result, the rate constant (k) can be obtained by fitting the change in Δf with time, as shown in Figures 2a and 3a, using the following equation (Figure S5, Supporting Information):

$$\Delta f = \Delta f_{\infty} (1 - e^{-kt}) \quad (3)$$

where Δf and Δf_{∞} are the frequency shifts at times t and ∞ , respectively. As can be seen from Figure 4a,b, the rate constant increases following the trends $\text{Na}^+ < \text{Mg}^{2+} < \text{Ca}^{2+} < \text{K}^+$ and $\text{Cl}^- \approx \text{SO}_4^{2-} < \text{Br}^- < \text{HCO}_3^-$ for cations and anions, respectively. This means that the effectiveness of enzyme to degrade the PCLPU film increases following the same ion trends.

It is expected that the specific ion effects on the enzymatic degradation of the PCLPU film should be closely correlated with the ion-specific enzymatic hydrolysis of the ester of PCLPU. To test this hypothesis, pNPB is employed as a model substrate to mimic the enzymatic hydrolysis of the ester of PCLPU, which can be described by the following equation:⁴⁰



where E, S, ES, k_1 , k_{-1} , k_{cat} , and P are the enzyme, substrate, enzyme–substrate complex, forward rate constant, reverse rate constant, catalytic rate constant, and product, respectively. As a result, k_{cat} can be obtained by curve fitting of the experimental data using the following equation (Figure S6, Supporting Information):⁴⁰

$$v_0 = \frac{k_{\text{cat}}[\text{E}]_t[\text{S}]}{K_M + [\text{S}]} \quad (5)$$

where v_0 , $[\text{E}]_t$, $[\text{S}]$, and K_M are the initial velocity, total enzyme concentration, substrate concentration, and Michaelis constant, respectively. Figure 4 shows that k_{cat} increases following the trends $\text{Na}^+ < \text{Mg}^{2+} < \text{Ca}^{2+} < \text{K}^+$ and $\text{Cl}^- \approx \text{SO}_4^{2-} < \text{Br}^- < \text{HCO}_3^-$ for cations and anions, respectively. Obviously, the effectiveness of enzyme in the enzymatic hydrolysis of pNPB increases following the same ion trends as those for the enzymatic degradation of the PCLPU film. Moreover, the relationship between k and k_{cat} is almost linear for both cations and anions. These facts suggest that the specific ion effects on the enzymatic degradation of the PCLPU film are closely correlated with the ion-specific enzymatic hydrolysis of the ester of PCLPU. In other words, the different ions present in the natural seawater have distinct capabilities to affect the enzymatic degradation of polymeric antibiofouling materials. Nevertheless, the different concentrations of the ions in the natural seawater may interfere with the observed specific ion effects. To further extract the specific ion effects while avoiding the influence of ionic strength, we have performed the studies on the enzymatic degradation of the PCLPU film in the presence of different types of ions at the same ionic strength as

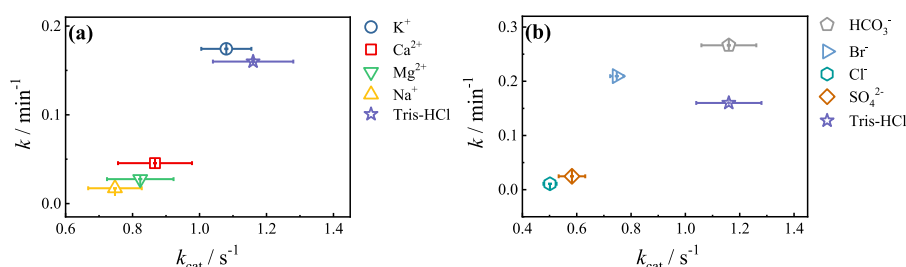


Figure 4. (a) Relationship between k and k_{cat} for the enzymatic degradation of the PCLPU film in the presence of different cations with Cl⁻ as the common anion. (b) Relationship between k and k_{cat} for the enzymatic degradation of the PCLPU film in the presence of different anions with Na⁺ as the common cation. Here, both the cations in (a) and the anions in (b) have the same concentrations as those in the natural seawater. The k and k_{cat} for the enzymatic degradation of the PCLPU film in the Tris-HCl buffer are also plotted for comparison, and the error bars represent the standard errors for the curve fitting of the experimental data.

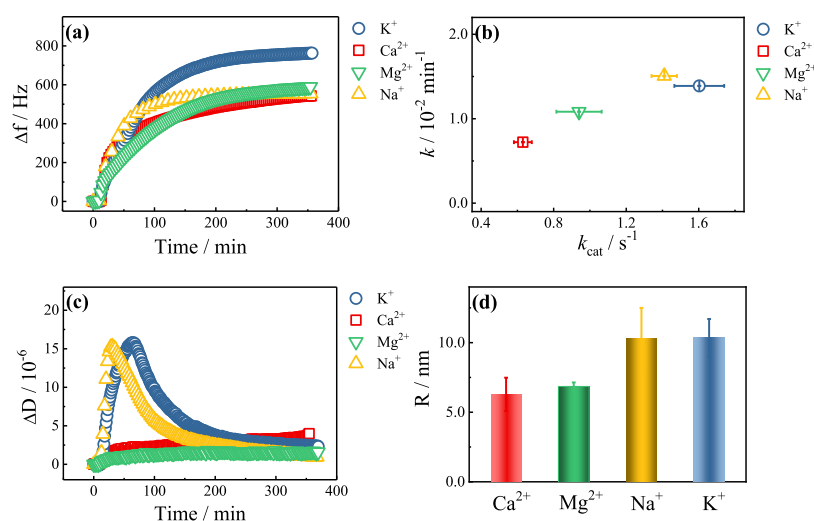


Figure 5. (a) Shift in frequency (Δf) for the enzymatic degradation of the PCLPU film as a function of time in the presence of different cations with Cl⁻ as the common anion. (b) Relationship between k and k_{cat} for the enzymatic degradation of the PCLPU film in the presence of different cations with Cl⁻ as the common anion. Here, the error bars represent the standard errors for the curve fitting of the experimental data. (c) Shift in dissipation (ΔD) for the enzymatic degradation of the PCLPU film as a function of time in the presence of different cations with Cl⁻ as the common anion. (d) RMS surface roughness (R) of the PCLPU film after a certain time of enzymatic degradation in the presence of different cations with Cl⁻ as the common anion. For Ca²⁺ and Mg²⁺, the surface roughness is determined at the time of enzymatic degradation of ~ 20 min. For K⁺ and Na⁺, the surface roughness is determined at the times of enzymatic degradation of ~ 55 and ~ 20 min, respectively, where ΔD has the maximum values. Here, the ionic strength is fixed at 0.72 M for all of the salt solutions.

that of the natural seawater (i.e., 0.72 M), as shown in Figures 5 and 6.

Figure 5a shows the time dependence of Δf for the enzymatic degradation of the PCLPU film in the presence of different cations at an ionic strength of 0.72 M. In Figure 5b, the values of k obtained by fitting the time dependence of Δf in Figure 5a using eq 3 (Figure S5, Supporting Information) show that the rate of degradation of the PCLPU film in the presence of different cations increases following the trend Ca²⁺ < Mg²⁺ < Na⁺ $\approx$ K⁺. The fact that k has an almost linear relationship with k_{cat} further suggests that the specific cation effects on the enzymatic degradation of the PCLPU film are closely correlated with the cation-specific enzymatic hydrolysis of the ester. The lipase (pI ~ 5.0)⁴¹ employed here is negatively charged in the Tris-HCl buffer (pH ~ 8.2). As a consequence, the different cations would specifically interact with the negatively charged carboxyl groups associated with the enzyme molecules according to the law of water matching affinities.^{42,43} More specifically, the more strongly hydrated divalent cations (e.g., Mg²⁺ and Ca²⁺) would exhibit stronger interactions with the strongly hydrated carboxyl groups than

the relatively weakly hydrated monovalent cations (e.g., Na⁺ and K⁺). The stronger interactions between the divalent cations and the carboxyl groups may generate a stronger disturbance on the enzyme structure, resulting in a lower enzyme activity.^{42,43} Thus, the enzyme activity in the presence of Na⁺ and K⁺ is higher than that in the presence of Mg²⁺ and Ca²⁺. A similar enzyme activity observed between Na⁺ and K⁺ may be related to the similar interactions between these two types of cations and the negatively charged protein molecules.⁴⁴ On the other hand, it is thought that a more kosmotropic ion gives rise to a higher enzyme activity due to an increase in affinity between enzyme and substrate, induced by the salting-out effect.^{15,45} This accounts for the higher enzyme activity in the presence of Mg²⁺ compared to that in the presence of Ca²⁺, since the former is considered as a more kosmotropic cation than the latter.

In Figure 5c, ΔD increases first and then decreases with increasing time for K⁺ and Na⁺, whereas ΔD only weakly changes with time for Ca²⁺ and Mg²⁺. In the presence of either K⁺ or Na⁺ at 0.72 M, the appearance of the maximum ΔD with time can be understood on the basis of the fact that the

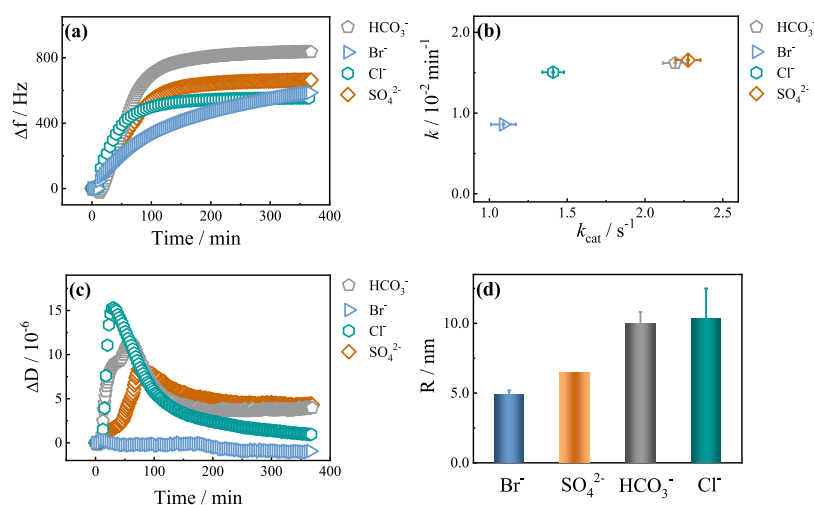


Figure 6. (a) Shift in frequency (Δf) for the enzymatic degradation of the PCLPU film as a function of time in the presence of different anions with Na^+ as the common cation. (b) Relationship between k and k_{cat} for the enzymatic degradation of the PCLPU film in the presence of different anions with Na^+ as the common cation. Here, the error bars represent the standard errors for the curve fitting of the experimental data. (c) Shift in dissipation (ΔD) for the enzymatic degradation of the PCLPU film as a function of time in the presence of different anions with Na^+ as the common cation. (d) RMS surface roughness (R) of the PCLPU film after a certain time of enzymatic degradation in the presence of different anions with Na^+ as the common cation. For Br^- , the surface roughness is determined at the time of enzymatic degradation of ~ 40 min. For Cl^- , HCO_3^- , and SO_4^{2-} , the surface roughness is determined at the times of enzymatic degradation of ~ 20 , ~ 50 , and ~ 65 min, respectively, where ΔD has the maximum values. Here, the ionic strength is fixed at 0.72 M for all of the salt solutions.

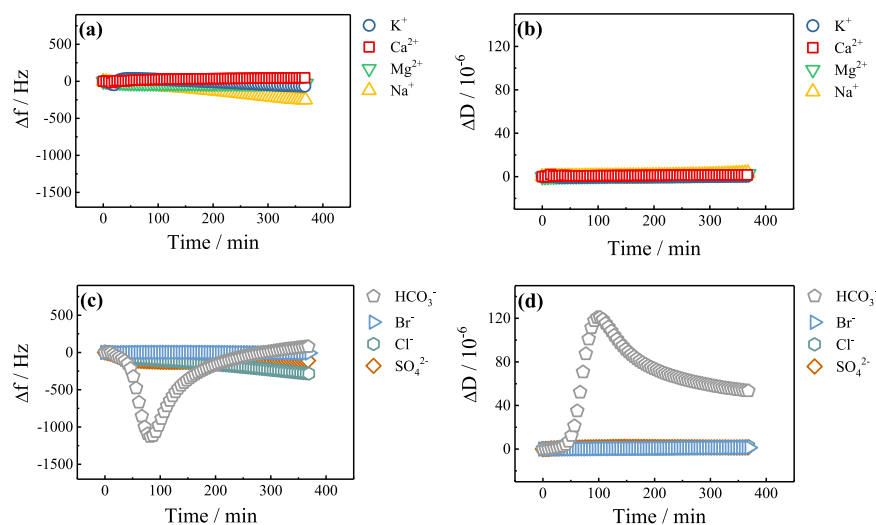


Figure 7. (a) Shift in frequency (Δf) for the enzymatic degradation of the PTIO film as a function of time in the presence of different cations with Cl^- as the common anion. (b) Shift in dissipation (ΔD) for the enzymatic degradation of the PTIO film as a function of time in the presence of different cations with Cl^- as the common anion. (c) Shift in frequency (Δf) for the enzymatic degradation of the PTIO film as a function of time in the presence of different anions with Na^+ as the common cation. (d) Shift in dissipation (ΔD) for the enzymatic degradation of the PTIO film as a function of time in the presence of different anions with Na^+ as the common cation. Here, both the cations in (a) and (b) and the anions in (c) and (d) have the same concentrations as those in the natural seawater.

degradation of PCLPU film in the presence of either K^+ or Na^+ at 0.72 M has a comparable rate to (Figure S7, Supporting Information) but a larger surface roughness than (Figure 5d) that in the presence of 468.0 mM Na^+ (Figure 2). In the presence of either Ca^{2+} or Mg^{2+} , the weak change in ΔD with time indicates that the influence of the degradation of PCLPU film on ΔD is largely offset by the effect generated by an increase in surface roughness of the PCLPU film during the degradation.

Figure 6a shows the time dependence of Δf for the enzymatic degradation of the PCLPU film in the presence of different anions at an ionic strength of 0.72 M. In Figure 6b,

the values of k obtained by fitting the time dependence of Δf in Figure 6a using eq 3 (Figure S5, Supporting Information) show that the rate of degradation of the PCLPU film in the presence of different anions increases following the trend $\text{Br}^- < \text{Cl}^- < \text{HCO}_3^- \approx \text{SO}_4^{2-}$. Again, the almost linear relationship between k and k_{cat} suggests that the specific anion effects on the degradation of the PCLPU film are closely related to the anion-specific enzymatic hydrolysis of the ester. The enzyme activity is higher for SO_4^{2-} and HCO_3^- compared to that for Cl^- and Br^- . This is because SO_4^{2-} and HCO_3^- are strongly hydrated kosmotropes and they can increase the enzyme activity by strengthening the affinity between enzyme and

substrate via the salting-out effect.¹⁵ In the classical Hofmeister series, Br^- is a chaotropic anion and Cl^- locates at the border between chaotropes and kosmotropes.^{46–55} Therefore, the direct binding of chaotropic anions to enzyme molecules would denature the enzyme conformation through the salting-in effect, leading to a decrease in enzyme activity.¹⁵ This explains the lower effectiveness of enzyme in the presence of Br^- than that in the presence of Cl^- . Likewise, the distinct changes in ΔD with time for the different anions shown in Figure 6c can be understood on the basis of the similar mechanism for the cations, as discussed in Figure 5c,d, by taking into account the surface roughness of the PCLPU film in the presence of different anions (Figure 6d).

It has been recently found that the combination of both degradable and hydrolyzable parts into one polymer provides an excellent antifouling performance in the marine field test even under static conditions.^{21,56} For example, the PTIO employed here has both the degradable ester part and the hydrolyzable silyl ester part, providing dual sources for the self-polishing properties of the PTIO film.²¹ During the time scale of our experiments, no obvious hydrolysis of the PTIO film can be observed in both the Tris-HCl buffer and the artificial seawater in the absence of lipase (Figure S8, Supporting Information) and also no obvious enzymatic hydrolysis of the silyl ester can be observed (Figure S9, Supporting Information); thus, here, we only focus on the specific ion effects on the enzymatic degradation of the PTIO film through the enzymatic hydrolysis of the ester. In Figure 7a,b, only small changes in Δf and ΔD can be observed as the time increases, implying that the PTIO film cannot be degraded in the presence of different cations with the same concentrations as those in the natural seawater. Also, no obvious degradation of the PTIO film can be observed in the presence of different anions with an exception of HCO_3^- , as shown in Figure 7c,d.

For the case of HCO_3^- , Δf decreases first and then increases with increasing time. Meanwhile, ΔD increases first and then decreases as time increases. The changes in Δf and ΔD in the presence of HCO_3^- could be induced by the formation of inhomogeneous structures within the PTIO film during the enzymatic degradation (Figures S10 and S11, Supporting Information). As discussed above, the HCO_3^- can give rise to a high enzyme activity, leading to enzymatic hydrolysis of the ester part of the PTIO film, while the silyl ester part of the PTIO film is kept intact in the presence of lipase. As a result, some inhomogeneous structures would be formed in the PTIO film with an increase in surface roughness. Upon the degradation, the solvent molecules would be trapped within these inhomogeneous structures, generating a large decrease in Δf and a significant increase in ΔD with increasing time in the initial stage, leading to the appearance of the minimum Δf and the maximum ΔD .

When the ionic strength of the salt concentrations is increased to 0.72 M, the PTIO film can be degraded by the enzyme in the presence of either K^+ or Na^+ , but no obvious degradation of the PTIO film can be observed in the presence of either Mg^{2+} or Ca^{2+} , as shown in Figure 8a. This is because, the enzyme has a higher activity in the presence of K^+ or Na^+ compared to that in the presence of Mg^{2+} or Ca^{2+} , as discussed in Figure 5. Nevertheless, K^+ and Na^+ lead to a different change in Δf during the degradation of PTIO film. Δf increases with increasing time in the presence of K^+ , whereas Δf rapidly decreases first and then increases with increasing time in the presence of Na^+ . On the other hand, ΔD sharply increases with

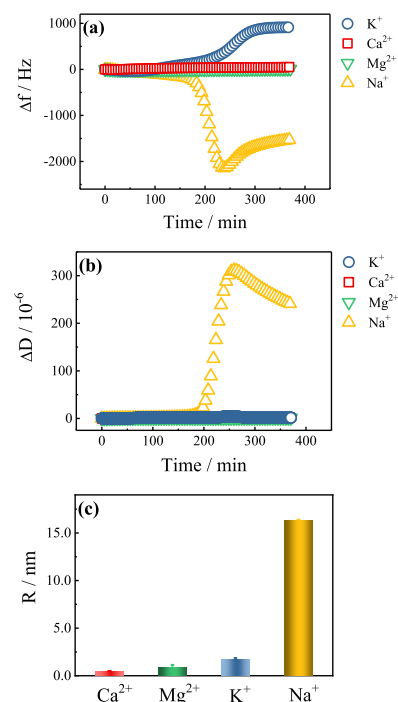


Figure 8. (a) Shift in frequency (Δf) for the enzymatic degradation of the PTIO film as a function of time in the presence of different cations with Cl^- as the common anion. (b) Shift in dissipation (ΔD) for the enzymatic degradation of the PTIO film as a function of time in the presence of different cations with Cl^- as the common anion. (c) RMS surface roughness (R) of the PTIO film after a certain time of enzymatic degradation in the presence of different cations with Cl^- as the common anion. For Ca^{2+} and Mg^{2+} , the surface roughness is determined at the time of enzymatic degradation of ~ 250 min. For K^+ and Na^+ , the surface roughness is determined at the times of enzymatic degradation of ~ 245 and ~ 250 min, respectively, where ΔD has the maximum values. Here, the ionic strength is fixed at ~ 0.72 M for all of the salt solutions.

increasing time until the maximum value is reached, followed by a decrease in ΔD with a further increase in time in the presence of Na^+ (Figure 8b). For K^+ , ΔD exhibits a similar change to that for Na^+ but in a much weaker extent (Figure S12, Supporting Information), which could be explained by an increase in surface roughness of the PTIO film during the degradation (Figure S13, Supporting Information). For the case of Na^+ , the appearance of both the minimum Δf and the maximum ΔD in remarkable values is indicative of the formation of inhomogeneous structures in the PTIO film with an increase in surface roughness during the degradation (Figure 8c). This is similar to the degradation of PTIO film in the presence of HCO_3^- , as shown in Figure 7c,d.

In the presence of SO_4^{2-} at 0.72 M, an increase in Δf with increasing time implies a degradation of PTIO film (Figure 9a). This should be attributed to the high enzyme activity in the presence of SO_4^{2-} , similar to that observed in Figure 6. On the other hand, as time increases, ΔD increases first and then decreases in the presence of SO_4^{2-} (Figure S14, Supporting Information), which may be due to a small increase in surface roughness of the PTIO film during the degradation (Figure S15, Supporting Information). For HCO_3^- and Cl^- , the appearance of both the minimum Δf (Figure 9a) and the maximum ΔD (Figure 9b) in remarkable values could be attributed to the formation of inhomogeneous structures in the

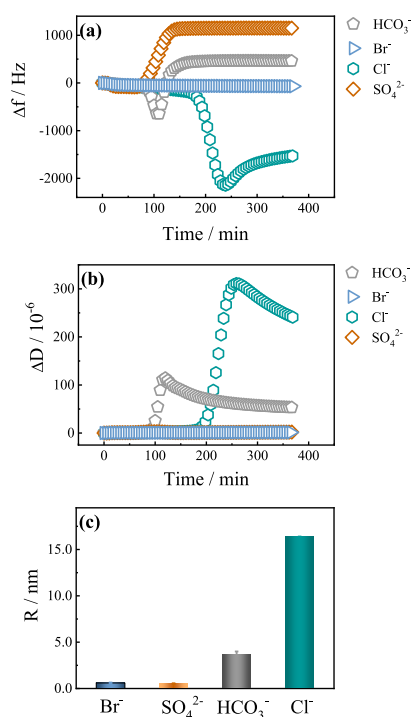


Figure 9. (a) Shift in frequency (Δf) for the enzymatic degradation of the PTIO film as a function of time in the presence of different anions with Na^+ as the common cation. (b) Shift in dissipation (ΔD) for the enzymatic degradation of the PTIO film as a function of time in the presence of different anions with Na^+ as the common cation. (c) RMS surface roughness (R) of the PTIO film after a certain time of enzymatic degradation in the presence of different anions with Na^+ as the common cation. For Br^- , the surface roughness is determined at the time of enzymatic degradation of ~ 250 min. For Cl^- , HCO_3^- , and SO_4^{2-} , the surface roughness is determined at the times of enzymatic degradation of ~ 250 , ~ 110 , and ~ 90 min, respectively, where ΔD has the maximum values. Here, the ionic strength is fixed at 0.72 M for all of the salt solutions.

PTIO film with increases in surface roughness during the degradation (Figure 9c), consistent with the observations for Na^+ shown in Figure 8. Additionally, no obvious enzymatic degradation of the PTIO film can be observed for Br^- at an ionic strength of 0.72 M because of the low enzyme activity in the presence of chaotropic Br^- , as discussed in detail in Figure 6.

CONCLUSIONS

We have investigated the specific ion effects on the enzymatic degradation of two types of polymeric marine antibiofouling materials. The observed ion specificities in the enzymatic degradation of the polymeric antibiofouling materials are closely correlated with the ion-specific enzymatic hydrolysis of the ester. Our study shows that the rough structures can be formed in the polymer films during the enzymatic degradation. The cation-specific enzymatic degradation of the polymeric antibiofouling materials is dominated by the direct specific interactions between the cations and the negatively charged enzyme molecules. For the anion-specific degradation, the kosmotropic anions increase the enzyme activity by increasing the affinity between the enzyme and substrate through the salting-out effect, whereas the chaotropic anions lead to a decrease in enzyme activity via the salting-in effect. The detailed mechanisms elucidated here will open up oppor-

tunities for the application of specific ion effects to modulate the biodegradation of polymeric antibiofouling materials in the marine environment.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.langmuir.9b01740.

pH values of salt solutions, results of control experiments, AFM results, curve fitting of experimental data, and other additional data (PDF)

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The authors declare no competing financial interest.

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REFERENCES

- (1) Lejars, M.; Margaillan, A.; Bressy, C. Fouling Release Coatings: A Nontoxic Alternative to Biocidal Antifouling Coatings. *Chem. Rev.* **2012**, *112*, 4347–4390.
- (2) Callow, J. A.; Callow, M. E. Trends in the Development of Environmentally Friendly Fouling-Resistant Marine Coatings. *Nat. Commun.* **2011**, *2*, No. 244.
- (3) Buskens, P.; Wouters, M.; Rentrop, C.; Vroon, Z. A Brief Review of Environmentally Benign Antifouling and Foul-Release Coatings for Marine Applications. *J. Colloid Interface Sci.* **2012**, *10*, 29–36.
- (4) Almeida, E.; Diamantino, T. C.; de Sousa, O. Marine Paints: The Particular Case of Antifouling Paints. *Prog. Org. Coat.* **2007**, *59*, 2–20.
- (5) Bressy, C.; Margaillan, A.; Faÿ, F.; Linossier, I.; Réhel, K. Tin-Free Self-Polishing Marine Antifouling Coatings. *Adv. Mar. Antifouling Coat. Technol.* **2009**, 445–491.
- (6) Chambers, L. D.; Stokes, K. R.; Walsh, F. C.; Wood, R. J. K. Modern Approaches to Marine Antifouling Coatings. *Surf. Coat. Technol.* **2006**, *201*, 3642–3652.
- (7) Dai, G.; Xie, Q.; Ma, C.; Zhang, G. Biodegradable Poly(ester-co-acrylate) with Antifoulant Pendant Groups for Marine Anti-Biofouling. *ACS Appl. Mater. Interfaces* **2019**, *11*, 11947–11953.
- (8) Carteau, D.; Vallée-Réhel, K.; Linossier, I.; Quiniou, F.; Davy, R.; Compère, C.; Delbury, M.; Faÿ, F. Development of Environmentally Friendly Antifouling Paints Using Biodegradable Polymer and Lower Toxic Substances. *Prog. Org. Coat.* **2014**, *77*, 485–493.
- (9) Xie, Q.; Pan, J.; Ma, C. F.; Zhang, G. Z. Dynamic Surface Antifouling: Mechanism and Systems. *Soft Matter* **2019**, *15*, 1087–1107.
- (10) Yu, J. Biodegradation-Based Polymer Surface Erosion and Surface Renewal for Foul-Release at Low Ship Speeds. *Biofouling* **2003**, *19*, 83–90.
- (11) Ma, J. L.; Ma, C. F.; Yang, Y.; Xu, W. T.; Zhang, G. Z. Biodegradable Polyurethane Carrying Antifoulants for Inhibition of Marine Biofouling. *Ind. Eng. Chem. Res.* **2014**, *53*, 12753–12759.

- (12) Xu, W. T.; Ma, C. F.; Ma, J. L.; Gan, T. S.; Zhang, G. Z. Marine Biofouling Resistance of Polyurethane with Biodegradation and Hydrolyzation. *ACS Appl. Mater. Interfaces* **2014**, *6*, 4017–4024.
- (13) Dickson, A. G.; Goyet, C. *Handbook of Methods for the Analysis of the Various Parameters of the Carbon Dioxide System in Sea Water*; Oak Ridge National Lab: Tennessee, 1994.
- (14) Millero, F. J.; Feistel, R.; Wright, D. G.; McDougall, T. J. The Composition of Standard Seawater and the Definition of the Reference-Composition Salinity Scale. *Deep Sea Res., Part I* **2008**, *55*, 50–72.
- (15) Bauduin, P.; Nohmie, F.; Touraud, D.; Neueder, R.; Kunz, W.; Ninham, B. W. Hofmeister Specific-Ion Effects on Enzyme Activity and Buffer pH: Horseradish Peroxidase in Citrate Buffer. *J. Mol. Liq.* **2006**, *123*, 14–19.
- (16) Salis, A.; Bilaničová, D.; Ninham, B. W.; Monduzzi, M. Hofmeister Effects in Enzymatic Activity Weak and Strong Electrolyte Influences on the Activity of *Candida rugosa* Lipase. *J. Phys. Chem. B* **2007**, *111*, 1149–1156.
- (17) Pokorná, J.; Heyda, J.; Konvalinka, J. Ion Specific Effects of Alkali Cations on the Catalytic Activity of HIV-1 Protease. *Faraday Discuss.* **2013**, *160*, 359–370.
- (18) Štěpánková, V.; Paterova, J.; Damborsky, J.; Jungwirth, P.; Chaloupkova, R.; Heyda, J. Cation-Specific Effects on Enzymatic Catalysis Driven by Interactions at the Tunnel Mouth. *J. Phys. Chem. B* **2013**, *117*, 6394–6402.
- (19) Yang, Z. Hofmeister Effects: an Explanation for the Impact of Ionic Liquids on Biocatalysis. *J. Biotechnol.* **2009**, *144*, 12–22.
- (20) Ma, C. F.; Zhang, W. P.; Zhang, G. Z.; Qian, P. Y. Environmentally Friendly Antifouling Coatings Based on Biodegradable Polymer and Natural Antifoulant. *ACS Sustainable Chem. Eng.* **2017**, *5*, 6304–6309.
- (21) Xie, Q. Y.; Ma, C. F.; Zhang, G. Z.; Bressy, C. Poly(ester)-Poly(silyl methacrylate) Copolymers Synthesis and Hydrolytic Degradation Kinetics. *Polym. Chem.* **2018**, *9*, 1448–1454.
- (22) Feller, G.; Thiry, M.; Arpigny, J. L.; Mergeay, M.; Gerday, C. Lipases From Psychrotropic Antarctic Bacteria. *FEMS Microbiol. Lett.* **1990**, *66*, 239–243.
- (23) Rodahl, M.; Höök, F.; Krozer, A.; Brzezinski, P.; Kasemo, B. Quartz Crystal Microbalance Setup for Frequency and Q-Factor Measurements in Gaseous and Liquid Environments. *Rev. Sci. Instrum.* **1995**, *66*, 3924–3930.
- (24) Höök, F.; Kasemo, B. Variations in Coupled Water, Viscoelastic Properties, and Film Thickness of a Mefp-1 Protein Film during Adsorption and Cross-linking: A Quartz Crystal Microbalance with Dissipation Monitoring, Ellipsometry, and Surface Plasmon Resonance Study. *Anal. Chem.* **2001**, *73*, 5796–5804.
- (25) Sauerbery, G. Verwendung von Schwingquarzen zur Wägung dünner Schichten und zur Mikrowägung. *Z. Phys.* **1959**, *155*, 206–222.
- (26) Atashin, S.; Pakshir, M.; Yazdani, A. Standard Practice for the Preparation of Substitute Ocean Water. *Trends Appl. Sci. Res.* **2008**, *5*, 61–67.
- (27) Lu, Y. P.; Zhang, X. H.; Fan, Z. Q.; Du, B. Y. Adsorption of PNIPAm₁₁₀-PEO₁₀₀-PPO₆₅-PEO₁₀₀-PNIPAm₁₁₀ Pentablock Terpolymer on Hydrophobic Gold. *Polymer* **2012**, *53*, 3791–3801.
- (28) Chen, T. Q.; Lu, Y. P.; Chen, T. Y.; Zhang, X. H.; Du, B. Y. Adsorption of PNIPAm_x-PEO₂₀-PPO₇₀-PEO₂₀-PNIPAm_x Pentablock Terpolymer on Gold Surfaces: Effects of Concentration, Temperature, Block Length, and Surface Properties. *Phys. Chem. Chem. Phys.* **2014**, *16*, 5536–5544.
- (29) Wang, T.; Long, Y. C.; Liu, L. D.; Wang, X.; Craig, V. S.; Zhang, G. Z.; Liu, G. M. Cation-Specific Conformational Behavior of Polyelectrolyte Brushes: From Aqueous to Nonaqueous Solvent. *Langmuir* **2014**, *30*, 12850–12859.
- (30) Kou, R.; Zhang, J.; Wang, T.; Liu, G. M. Interactions between Polyelectrolyte Brushes and Hofmeister Ions: Chaotropes versus Kosmotropes. *Langmuir* **2015**, *31*, 10461–10468.
- (31) Wang, T.; Kou, R.; Liu, H. L.; Liu, L. D.; Zhang, G. Z.; Liu, G. M. Anion Specificity of Polyzwitterionic Brushes with Different Carbon Spacer Lengths and Its Application for Controlling Protein Adsorption. *Langmuir* **2016**, *32*, 2698–2707.
- (32) Wang, T.; Wang, X. W.; Long, Y. C.; Liu, G. M.; Zhang, G. Z. Ion-Specific Conformational Behavior of Polyzwitterionic Brushes: Exploiting It for Protein Adsorption/Desorption Control. *Langmuir* **2013**, *29*, 6588–6596.
- (33) Rechendorff, K.; Hovgaard, M. B.; Foss, M.; Besenbacher, F. Influence of Surface Roughness on Quartz Crystal Microbalance Measurements in Liquids. *J. Appl. Phys.* **2007**, *101*, No. 114502.
- (34) Turon, X.; Rojas, O. J.; Deinhammer, R. S. Enzymatic Kinetics of Cellulose Hydrolysis: A QCM-D Study. *Langmuir* **2008**, *24*, 3880–3887.
- (35) Ohlsson, G.; Langhammer, C.; Zorić, I.; Kasemo, B. A Nanocell for Quartz Crystal Microbalance and Quartz Crystal Microbalance with Dissipation-Monitoring Sensing. *Rev. Sci. Instrum.* **2009**, *80*, No. 083905.
- (36) Urbakh, M.; Daikhin, L. Roughness Effect on the Frequency of a Quartz-Crystal Resonator in Contact with A Liquid. *Phys. Rev. B* **1994**, *49*, 4866–4870.
- (37) Hou, Y.; Chen, J.; Sun, P.; Gan, Z. H.; Zhang, G. Z. In Situ Investigations on Enzymatic Degradation of Poly(ϵ -caprolactone). *Polymer* **2007**, *48*, 6348–6353.
- (38) Lam, H.; Gong, X.; Wu, C. Novel Differential Refractometry Study of the Enzymatic Degradation Kinetics of Poly(ethylene oxide)-*b*-Poly(epsilon-caprolactone) Particles Dispersed in Water. *J. Phys. Chem. B* **2007**, *111*, 1531–1535.
- (39) Nie, T.; Zhao, Y.; Xie, Z.; Wu, C. Micellar Formation of Poly(caprolactone-*block*-ethylene oxide-*block*-caprolactone) and Its Enzymatic Biodegradation in Aqueous Dispersion. *Macromolecules* **2003**, *36*, 8825–8829.
- (40) Balcells, C.; Pastor, I.; Vilaseca, E.; Madurga, S.; Cascante, M.; Mas, F. Macromolecular Crowding Effect upon In Vitro Enzyme Kinetics: Mixed Activation-Diffusion Control of the Oxidation of NADH by Pyruvate Catalyzed by Lactate Dehydrogenase. *J. Phys. Chem. B* **2014**, *118*, 4062–4068.
- (41) Padilha, G. d. S.; Santana, J. C. C.; Alegre, R. M.; Tambourgi, E. B. Extraction of Lipase from *Burkholderia cepacia* by PEG/Phosphate ATPS and Its Biochemical Characterization. *Braz. Arch. Biol. Technol.* **2012**, *55*, 7–19.
- (42) Collins, K. D. Ions from the Hofmeister Series and Osmolytes: Effects on Proteins in Solution and in the Crystallization Process. *Methods* **2004**, *34*, 300–311.
- (43) Collins, K. D. Ion Hydration: Implications for Cellular Function, Polyelectrolytes, and Protein Crystallization. *Biophys. Chem.* **2006**, *119*, 271–281.
- (44) Wang, X. W.; Liu, G. M.; Zhang, G. Z. Effect of Surface Wettability on Ion-Specific Protein Adsorption. *Langmuir* **2012**, *28*, 14642–14653.
- (45) Wondrak, E. M.; Louis, J. M.; Stephen, O. The Effect of Salt on the Michaelis-Menten Constant of the HIV-1 Protease Correlates with the Hofmeister Series. *FEBS Lett.* **1991**, *280*, 344–346.
- (46) Lo Nostro, P.; Ninham, B. W. Hofmeister Phenomena: An Update on Ion Specificity in Biology. *Chem. Rev.* **2012**, *112*, 2286–2322.
- (47) Parsons, D. F.; Bostrom, M.; Lo Nostro, P.; Ninham, B. W. Hofmeister Effects: Interplay of Hydration, Nonelectrostatic Potentials, and Ion Size. *Phys. Chem. Chem. Phys.* **2011**, *13*, 12352–12367.
- (48) Kunz, W. Specific Ion Effects in Colloidal and Biological Systems. *Curr. Opin. Colloid Interface Sci.* **2010**, *15*, 34–39.
- (49) Mazzini, V.; Craig, V. S. Specific-Ion Effects in Non-Aqueous Systems. *Curr. Opin. Colloid Interface Sci.* **2016**, *23*, 82–93.
- (50) Medda, L.; Barse, B.; Cugia, F.; Boström, M.; Parsons, D. F.; Ninham, B. W.; Monduzzi, M.; Salis, A. Hofmeister Challenges: Ion Binding and Charge of the BSA Protein as Explicit Examples. *Langmuir* **2012**, *28*, 16355–16363.
- (51) Salis, A.; Ninham, B. W. Models and Mechanisms of Hofmeister Effects in Electrolyte Solutions, and Colloid and Protein Systems Revisited. *Chem. Soc. Rev.* **2014**, *43*, 7358–7377.

(52) Mazzini, V.; Craig, V. S. What is the Fundamental Ion-Specific Series for Anions and Cations? Ion Specificity in Standard Partial Molar Volumes of Electrolytes and Electrostriction in Water and Non-Aqueous Solvents. *Chem. Sci.* **2017**, *8*, 7052–7065.

(53) Zhang, Y.; Cremer, P. S. Chemistry of Hofmeister Anions and Osmolytes. *Annu. Rev. Phys. Chem.* **2010**, *61*, 63–83.

(54) Chen, X.; Flores, S. C.; Lim, S.-M.; Zhang, Y.; Yang, T.; Kherb, J.; Cremer, P. S. Specific Anion Effects on Water Structure Adjacent to Protein Monolayers. *Langmuir* **2010**, *26*, 16447–16454.

(55) Salis, A.; Cugia, F.; Parsons, D. F.; Ninham, B. W.; Monduzzi, M. Hofmeister Series Reversal for Lysozyme by Change in pH and Salt Concentration: Insights from Electrophoretic Mobility Measurements. *Phys. Chem. Chem. Phys.* **2012**, *14*, 4343–4346.

(56) Zhou, X.; Xie, Q. Y.; Ma, C. F.; Chen, Z. J.; Zhang, G. Z. Inhibition of Marine Biofouling by Use of Degradable and Hydrolyzable Silyl Acrylate Copolymer. *Ind. Eng. Chem. Res.* **2015**, *54*, 9559–9565.